

Prototype-Based Modality-Specific Feature Distillation for Incomplete Multimodal Sentiment Analysis

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Abstract—Incomplete multimodal sentiment analysis is to recognize human emotional states by leveraging the incomplete multimodal information including text, audio, and visual data. Existing methods mainly focus on reconstructing from available modalities, while ignoring the specific information of missing modalities and the inconsistent distribution between original and recovered data. In this work, we propose a novel Prototype-based modality-specific feature Distillation Network called ProDNet, which can recover the specific information of missing modalities with more consistent distributions between original and recovered data. In particular, we design a category-specific prototype for each modality to store the modality-specific information, and reconstruct the missing modality by combining the invariant features from existing modalities with the prototype of the missing modality in a feature self-distillation manner. To obtain favorable invariant features, we design an invariant feature distillation strategy, in which the weight-gating mechanism is introduced to enhance invariant feature representation. Finally, to reduce distributional inconsistency, we design a distribution consistency distillation loss to enforce the consistent distribution of original and recovered data by considering the sample correlation between teacher and student networks. Extensive experiments on two benchmark datasets, CMU-MOSI and CMU-MOSEI, demonstrate the superior performance of the proposed method, particularly in scenarios with higher missing rates.

Index Terms—Multimodal sentiment analysis, missing modalities, knowledge distillation.

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Code is available at <https://github.com/Yongbo-Wang/ProDNet>.

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I. INTRODUCTION

MULTIMODAL sentiment analysis (MSA) is an important branch of multimodal machine learning, intending to combine text, audio, and visual information in videos to recognize human emotional states [1], [2], [3], [4]. It has wide applications in fields such as healthcare [5] and robotics [6], [7]. However, in real-life scenarios, modalities are sometimes missing due to unavoidable circumstances. For instance, visual, acoustic, and text data may be ubiquitously missing caused by camera occlusion, ambient noise disruption, and inaccurate translation, respectively. When a modality is missing, models trained with the complete modality often experience significant performance degradation. The issue of missing modalities makes the MSA not well adapted to downstream tasks [8], [9], [10].

To address the problem of missing modalities, a typical approach is to use the available modalities to recover or reconstruct missing data. The core of this approach is to utilize the encoder-decoder structure to fully capture the feature information of the available modalities for reconstruction. Enhancing the robustness of joint multimodal representations through a deeper exploration of available modalities is an intuitive approach to handling missing modalities. Pham et al. [12] introduce a cross-modality translation strategy to learn robust joint multimodal representations. They incorporate a cyclic consistency loss to guarantee that the joint representation retains as much information as possible for all modalities. However, it is only to learn a more robust multimodal representation, while ignoring the information about the missing modalities. For the general case of missing modalities, Zeng et al. [13] propose a tag-assisted transformer encoder network. Introducing tags for different missing cases allows the network to focus better on the missing modalities. To further capture the time and speaker information in the conversation, Lian et al. [14] utilize graph neural networks to reconstruct the missing data. However, the information of each modality can be essentially classified into invariant features, which contain the commonalities of the modality, and specific features that are unique to each modality, such as sound pitch and loudness. These methods utilize the information of existing modalities to reconstruct the missing data. The reconstruction is carried out using the invariant features of the modalities without considering modality-specific information. Additionally, the absence of modality-specific features can also result

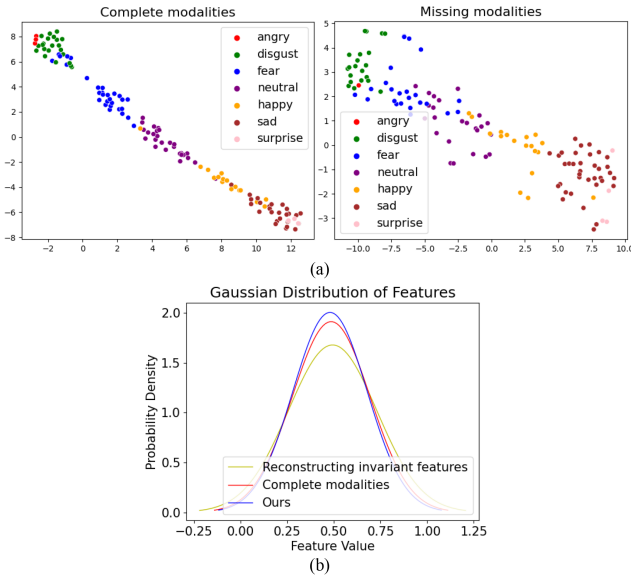


Fig. 1. Results on the CMU-MOSI [11] dataset. (a) The feature distributions of the complete and missing modalities on the results of the seven classifications. (b) Comparison of Gaussian distribution curves of features resulting from different methods. Specifically, the red curve depicts the Gaussian distribution without missing modalities, the yellow curve illustrates the scenario where invariant features of existing modalities are utilized to complement missing ones, and the blue curve represents the Gaussian distribution after complementing both invariant features and specific features, as well as ensuring relational consistency.

in inconsistencies in the distribution of the reconstructed data compared to the original data.

As shown in Fig. 1(a), the feature distribution of the complete modalities shows a compact bar, and each category can be better distinguished from the others. When missing modalities occur, the feature distribution becomes scattered. The phenomenon indicates that there is an inconsistency problem between the complete and missing modalities. Wang et al. [15] propose a shared-specific feature modeling approach to reconstruct missing modalities, demonstrating the importance of invariant features. Solely relying on invariant features between available modalities for completing missing ones results in a noticeable gap, as shown in the yellow and red curves in Fig. 1(b). Motivated by the widespread use of representing prototypes by computing the average feature vectors in the same class [16], we propose a prototype reconstruction self-distillation strategy to comprehensively reconstruct the missing modalities. We use category-specific prototypes for each modality to learn modality-specific information. Consequently, considering both invariant features and specific features, as shown in the blue curve in Fig. 1(b), our method fits the Gaussian curve, which is closer to the complete modalities. This evidences the significance of missing modality-specific features, offering a more comprehensive approach to complement missing modalities.

Other methods tend to generate missing modalities using a generative model [17], [18] to address the distribution inconsistency problem via employing normalized flow or diffusion models. However, generation-based methods demand a substantial amount of computational resources. Additionally, the inherent uncertainty of the generative approach may result in

wide fluctuations within the network [19]. In contrast, knowledge distillation offers a more efficient alternative. Transferring knowledge from a complex model to a simpler one facilitates enhanced generalization performance while minimizing computational overhead [20]. In the face of missing modalities, the teacher network can transmit information about the complete modality to the student network, thereby assisting the student network in reconstructing the missing modalities. Nevertheless, simultaneously distilling invariant features and specific features within a single teacher network poses challenges in balancing their importance effectively. Hence, we propose an invariant feature distillation strategy to learn invariant features from the teacher network, which is optimized using a weight-gating mechanism. We utilize the invariant feature and specific feature mutual orthogonalization strategy to guide the direction of the update. Additionally, we employ the self-distillation method to handle specific features. Meanwhile, recognizing that the student network may encounter distributional inconsistency problems arising from missing modalities, we design a distribution consistency distillation loss to impose constraints.

In summary, we propose a **Prototype-based modality-specific feature Distillation Network (ProDNet)** to address the aforementioned challenges. It mainly consists of a Prototype Reconstruction Self-Distillation (PRD) to enhance invariant and specific features via prototypes, an Invariant Feature Distillation (IFD) to guide the student network to learn comprehensive invariant features from the teacher network, and a Distribution Consistency Distillation (DCD) to ensure distribution consistency.

The major contributions of this paper could be summarized as follows:

- To comprehensively reconstruct the missing modalities, we propose a prototype reconstruction self-distillation method that considers specific and invariant features of the missing modalities. Specifically, we design category-specific prototypes to store modality-specific information and integrate the invariant features of available modalities for the missing modality recovery.
- To better learn the invariant features, we propose an invariant feature distillation strategy to transfer knowledge from the teacher network, followed by a weight-gate mechanism to further optimize the invariant features for the teacher network. In addition, we introduce invariant and specific feature orthogonality to constrain the update direction.
- To prevent the distributional inconsistency problem, we propose a distribution consistency distillation loss, which takes into account sample-to-sample relationships to reduce the distributional gap between the recovered modalities and the original modalities.

Comparative analysis for incomplete multimodal sentiment analysis is conducted on CMU-MOSI [11] and CMU-MOSEI [21] datasets. The results confirm the superiority of our proposed method.

II. RELATED WORK

In this section, we will briefly review topics related to incomplete multimodal learning and knowledge distillation.

A. Incomplete Multimodal Learning

To address the widespread issue of missing modalities in realistic scenarios, a diverse range of incomplete multimodal learning methods has been proposed. The methods can be roughly classified into two groups: modality-completion methods [12], [22], [23], [24] and modality-generation methods [18], [25], [26], [27]. These methods play a crucial role in handling incomplete multimodal data and have been designed to offer effective solutions for the challenges posed by missing modalities.

The modality-completion approach operates on the principle of estimating and reconstructing missing modalities based on available modality information. Pham et al. [12] investigate a translation strategy from source to target modalities to learn robust representations by acquiring all modality information through cyclic consistency loss. It is shown that the method can mitigate the effects of missing data to a certain extent. Therefore, Zhao et al. [28] apply cyclic consistency learning to the interpolation of missing modalities and design an imaginary module based on a cascaded residual autoencoder to reconstruct missing data. Yuan et al. [29] design a Transformer-based feature reconstruction approach, which utilizes Transformers to extract intra-modal and inter-modal relationships, and guides the recovery of missing modes through a feature reconstruction network. Lian et al. [14] use graph neural networks to capture temporal information in dialogues to reconstruct missing modalities. Previous methods are only for a certain domain, similar to classification or segmentation. Wang et al. [15] propose a shared-specific feature modeling approach to learn shared and specific features of different modalities to adapt to multiple tasks with different missing.

The modality-generation method uses generative models to learn new data that has a similar distribution to the original data. Shang et al. [27] treat each view as a separate domain and utilize the generative adversarial network (GAN) [30] to learn an equivalent mapping for each view and estimate the missing data. Wang et al. [17] propose to use the reversibility of normalized streams to generate missing data for the problem of inconsistent data distribution. Wang et al. [18] choose a score-based diffusion model for generation, while preventing semantic inconsistencies in the generated data by introducing available modalities as a supervisory condition. However, modality-completion methods do not take into account the specific features of the missing modality, and modality-generation methods require a large amount of computational resources, as well as a large uncertainty in the generated results.

B. Knowledge Distillation

Knowledge distillation seeks to convey extensive knowledge from the teacher network to the less powerful student network, aiding in the student network's learning process. Generally, it can be divided into three types according to the distillation method: response-based distillation that matches the output of the teacher and student networks [31], [32], feature-based distillation that transfers high-level feature knowledge from the teacher model to the student model [33], [34], [35], relationship-based distillation that focuses on the relationship of samples [36], [37].

Knowledge distillation has widespread applications in the domain of incomplete multimodal learning. The teacher network can be a comprehensive model trained on all modalities, while the student network focuses on the missing modalities [38]. Due to the different inputs of teacher and student networks in incomplete multimodal learning, direct feature transfer of knowledge may lead to overfitting [39]. In the missing modality task, the performance of the unified framework is often not as well as the customized framework. Wei et al. [20] improve the performance of the unified framework by focusing on the information of separate boundaries while proposing a modality-aware regularization algorithm to mine the weak modality combinations. However, these existing techniques only consider one aspect alone, using available modalities to recover missing modalities, or reducing the distribution problem through generation or distillation. Knowledge distillation can handle the missing modality problem more effectively. Therefore, we propose a unified approach that not only considers the information of the available modalities but also takes into account the distributional inconsistencies caused by the missing ones.

III. METHOD

In this paper, we focus on multimodal sentiment analysis under missing random. Our proposed framework, ProDNet, mainly consists of a teacher network with dual-stream feature decoupling and a student network that combines three distillation strategies, as shown in Fig. 2. The teacher network is trained using complete modalities, and the features are decoupled to obtain both invariant and specific features through a dual-stream structure. Additionally, the invariant features are further optimized using a weight-gating mechanism. For the student network, data with missing modalities are utilized for training. When a missing modality occurs, we employ Invariant Feature Distillation (IFD) to learn robust invariant features from available modalities, Prototype Reconstruction Self-Distillation (PRD) to capture specific features of the missing modality, combine invariant features with specific features to reconstruct the missing modality, and finally align the distribution of the student network with that of the teacher network through Distribution Consistency Distillation (DCD). Thus, considering both invariant and specific features of the missing modality with distributional alignment, the missing modality will be comprehensively reconstructed.

A. Problem Definition and Notations

The multimodal sentiment analysis task focuses on predicting the speaker's sentiment from a given video sequence S , which consists of three main modalities: visual v , text t , and audio a . Define the above three modalities as $S = [X_m \in \mathbb{R}^{T_m \times d_m}]$, $m \in \{a, v, t\}$, where T_m denotes the length of the sequence, d_m denotes the feature dimension, and m means different modalities. Assuming the audio modality is missing, the multimodal data with the missing modality can be defined as (X_a^*, X_v, X_t) . Thus, we can define the problem as described below, for a given triple of data (X_a, X_v, X_t) , randomly perform a missing operation on one or two of the modalities, and then

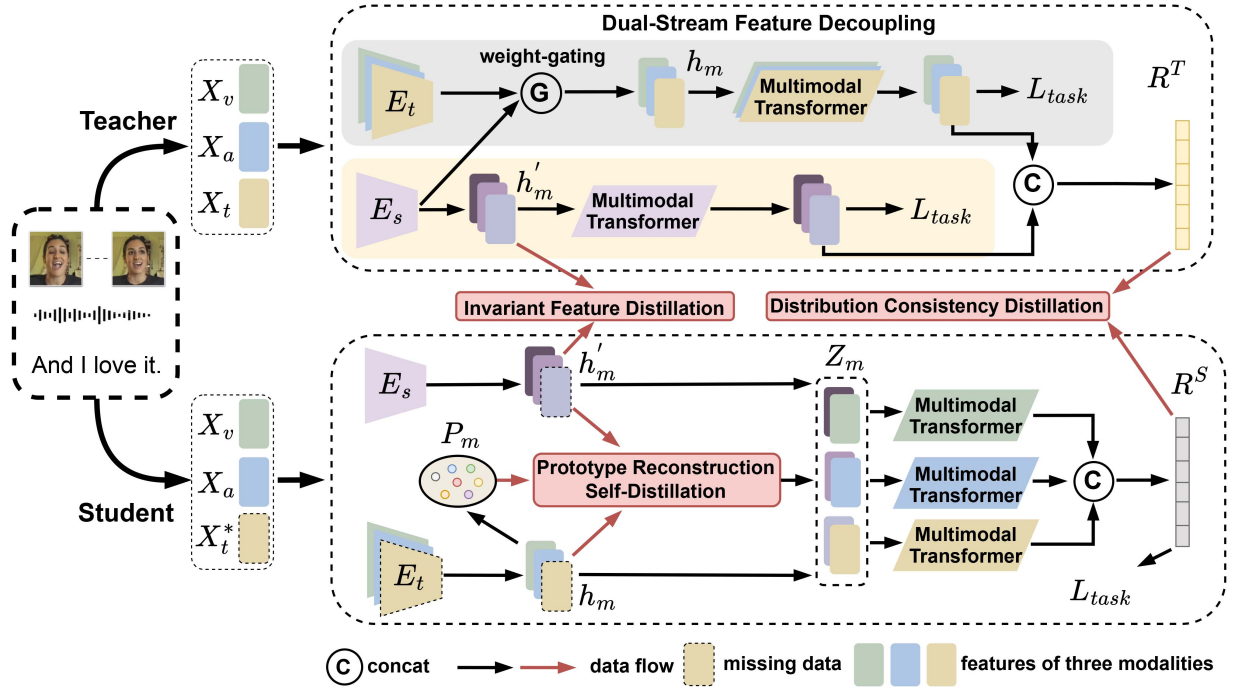


Fig. 2. Overview of our ProDNet method. The overall structure is a teacher-student network. The teacher network is a two-stream structure that learns a more robust multimodal representation through weight-gating, which utilizes the complete modality for training. Instead, the student network is trained with missing modalities, utilizing three distillation strategies to reconstruct the missing modalities comprehensively. Where E denotes the encoder, which extracts the shallow features of the modality. In the inference phase, we only utilize the student network to process the data.

analyze the speaker's emotions (positive, negative, etc.) based on the available modalities.

B. Dual-Stream Feature Decoupling of Teacher Network

Our proposed method is a distillation framework that consists of teacher and student networks. For the teacher network, we utilize dual-stream feature decoupling, as illustrated in Fig. 2. Decoupling multimodal data into invariant and specific features can improve the model's robustness and better exploit the sharing and specificity between different modalities, thereby fully utilizing the information from each modality. In our pursuit of achieving three-modality feature decoupling, we employ a dual-stream structure, where one branch learns invariant features through the shared feature encoder, and the other branch learns specific features through the specific feature encoder for each modality. To account for the different dimensions of various modalities, we utilize three independent 1D temporal convolutional layers to aggregate temporal information and obtain low-level multimodal features before passing through the encoder. Additionally, to facilitate subsequent feature decoupling, all modalities are scaled to the same feature dimension.

The shared feature encoder E_s , composed of a fully connected layer, maps information from different modalities into a common subspace under distributional constraints, aiming to learn the invariant features of the three modalities, denoted as (h'_a, h'_v, h'_t) . Specifically, these invariant features represent modality-agnostic embeddings that capture shared semantic information across modalities. The shared encoder E_s projects heterogeneous inputs into a unified latent space, enabling the

model to extract cross-modal representations that are independent of modality-specific characteristics. To achieve this, we employ the Central Moment Discrepancy (CMD) [40], which introduces a distance constraint to minimize discrepancies between modalities. By aligning the order-wise moment statistics of their feature distributions, CMD facilitates the effective learning of modality-invariant representations. We define this constraint as L_{CMD} and minimize it to obtain the invariant features.

$$L_{CMD} = \frac{1}{3} \sum_{(m_1, m_2)} CMD_K(h'_{m_1}, h'_{m_2}),$$

$$(m_1, m_2) \in \{(a, v), (a, t), (v, t)\}, \quad (1)$$

$$CMD_K(h'_{m_1}, h'_{m_2}) = \|E(h'_{m_1}) - E(h'_{m_2})\|_2$$

$$+ \sum_{k=2}^K \|C_k(h'_{m_1}) - C_k(h'_{m_2})\|_2, \quad (2)$$

where $C_k(X) = E((X - E(X))^k)$ is the vector of all k^{th} order sample central moments of the coordinates of X , and $E(X)$ is the empirical expectation vector of the sample X .

The specific feature encoder comprises three different encoders: audio encoder E_a , visual encoder E_v , and text encoder E_t . Each modality is encoded separately, and the resulting features are constrained using a task-related loss to extract sentiment-related features specific to each modality, denoted as (h_a, h_v, h_t) . Each encoder focuses on capturing unique cues of its own modality, such as facial expressions and motion patterns for the visual branch, tone and prosody for the audio branch,

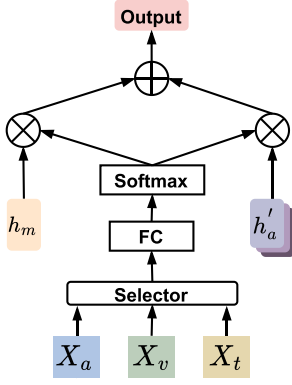


Fig. 3. Components of the weight-gating mechanism.

and emotional words for the textual branch. These encoders are optimized through the task-related loss L_{task} , ensuring that the learned features retain discriminative, modality-unique characteristics that complement the invariant subspace.

To enable the teacher network to better optimize invariant features, we design a weight-gating mechanism. This mechanism selectively fuses specific features with invariant features, thereby facilitating their further optimization. Compared to the independent two-stream network, this approach ensures that invariant features in a shared branch are influenced not only by the classification loss of that branch but also by each modality-specific branch. Consequently, this multiple optimization strategy allows the teacher network to learn more robust invariant features. As shown in Fig. 3, the gating structure relies on a single-layer feed-forward network that uses softmax as its activation function. It processes the input data from the three modalities to act as selectors and computes the weighted sum of the chosen vectors. The specific computational procedure is as follows:

$$G_m(\vec{X}, h_m) = W_m(\vec{X})I_m(h_m), m \in \{a, v, t\}, \quad (3)$$

where $\vec{X}$ denotes the three modality features, $W_m(\vec{X})$ is a weighting function that computes the weight vectors of the different features through a linear transformation and softmax layer, and $I_m(h_m)$ denotes the invariant features of the three modalities combined with the modality m . The calculation process is as follows:

$$W_m(\vec{X}) = \text{Softmax}(FC_m(\vec{X})), \quad (4)$$

$$I_m(h_m) = [h_m, h'_a, h'_v, h'_t]. \quad (5)$$

For both branches, after obtaining the invariant and specific features, they are fed into the corresponding multimodal transformer and subsequently concatenated to predict the sentiment category. We define the overall optimization goal of the teacher network as L_t :

$$L_t = L_{task} + \alpha L_{CMD}, \quad (6)$$

$$L_{task} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (7)$$

where α is the hyperparameter and N denotes the number of samples. While y_i and $\hat{y}_i$ denote the ground truth of samples and the label of the model prediction, respectively.

C. Prototype Reconstruction Self-Distillation: PRD

For the problem of missing modalities, modality completion is a widely used method. However, directly reconstructing the missing modality using the available modalities only captures the invariant features of the missing modality, while neglecting the modality-specific features. It will lead to an inconsistent distribution of the reconstructed and original data. Additionally, the invariant features of the available modalities contain crucial information about the missing modality, which is essential for effective complementation. For the student network, we reconstruct the missing modality in terms of both invariant and specific features.

Consequently, we propose the PRD strategy to consider both modality-invariant and modality-specific features. Specifically, we define a prototype $P_m = \{P_m^1, P_m^2, \dots, P_m^K\}$, $m \in \{a, v, t\}$ for each modality, where K is defined as the number of sentiment categories, and the other two modalities have the same prototype definition. During the training process, we keep updating and iterating on each prototype. The modality-specific features in the current batch are treated as new prototypes P_m^{new} . The old prototype P_m^{old} , which is updated from the previous round, is combined with the new prototype to get the current prototype P_m^{cur} . The computational procedure is as follows:

$$P_m^{new} = E_m(X_m), \quad (8)$$

$$P_m^{cur} = \eta P_m^{old} + (1 - \eta) P_m^{new}, \quad (9)$$

where η is the hyperparameter and h_m denotes modality-specific features of the student network.

In case of a missing modality, the weighted attention result of the missing modality prototype is computed based on the relationship between prototypes and samples of the available modalities. After that, the weighted attention is used to obtain the prototype combined with the invariant features of the available modality to recover the missing modality. As illustrated in Fig. 4(a), the attention is calculated as follows:

$$Att_m = \text{Softmax} \left(\frac{FC_S^m(E_m(X_m))(FC_P^m(P_m))^T}{K} \right), \quad (10)$$

where FC_S^m and FC_P^m are the two linear layers of samples and prototypes in m modality, K is the number of categories respectively, and E_m is the specific feature encoder of m modality in the student network.

After getting the attention of available modalities and prototypes, the missing modalities are reconstructed by combining the invariant features of available modalities. The reconstruction feature Z_m^* is computed as follows:

$$Z_m^* = h'_m + Att_m \times FC_P^*(P_m^*), \quad (11)$$

where P_m^* denotes the prototype of the missing modality, FC_P^* is a linear layer of the prototype of the missing modality, and h'_m denotes the invariant features of the available modalities.

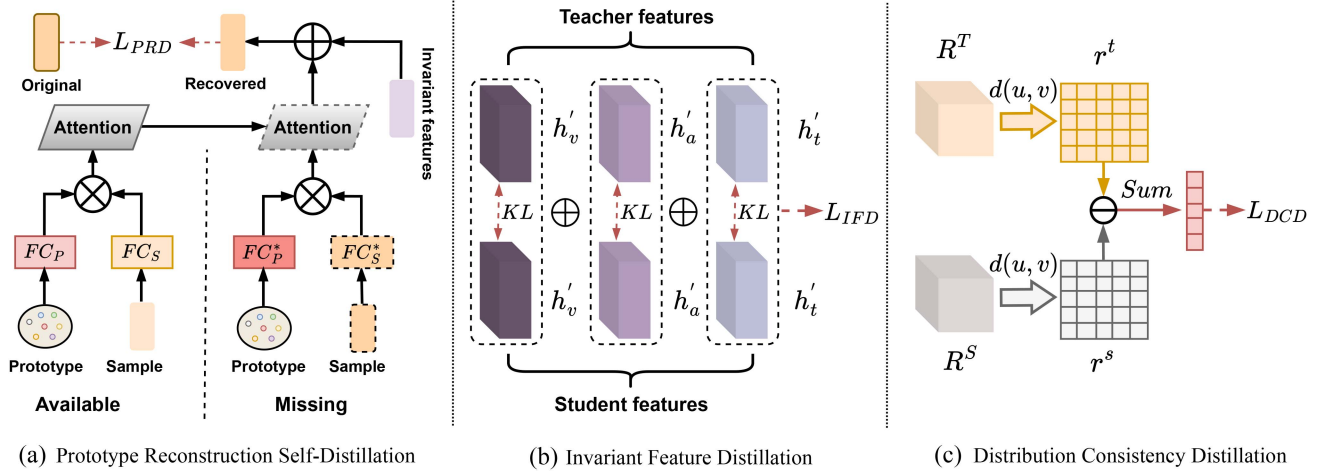


Fig. 4. Specific structure of the three distillation strategies. (a), (b), and (c) represent prototype reconstruction self-distillation, invariant feature distillation, and distribution consistency distillation, respectively.

When one modality is missing with two available modalities, the process of computing the attention is as follows:

$$Att = (Att_{m_1} + Att_{m_2})/2, \quad (12)$$

where m_1 and m_2 represent different modalities.

For different missing patterns, the recovered features can be represented as Z_m^* , and Z_m denotes the features of the original data. Consequently, the reconstruction loss L_{PRD} is defined as:

$$L_{PRD} = \sum \|Z_m^* - Z_m\|_F^2. \quad (13)$$

D. Invariant Feature Distillation: IFD

The modality-invariant features contain information about the three modalities as well as the missing modalities. To enhance the learning of invariant features, the teacher network will further optimize them through the weight-gating mechanism. Therefore, we propose an IFD strategy that aims to transfer robust invariant features from the teacher network to the student network, enhancing the student network's ability to reconstruct missing data subsequently. Furthermore, we orthogonalize the invariant and specific features in the student network to facilitate learning information in different directions.

Initially, we decouple the information of different modalities into invariant features and specific features. Assuming that one of the visual modalities is missing, we are unable to transfer the specific features of the visual modality to the student network because there is no visual modality in the student network. Conversely, modality invariant features can be well transferred and facilitate the distillation. With more robust invariant features, student networks can be driven to learn better information related to sentiment, facilitating subsequent prediction and reconstruction. To quantify the difference between the output probabilities of the two networks, we employ the Kullback-Leibler divergence (KL divergence) as a distance function, as in Fig. 4(b). Defining the invariant features of the teacher network and the student network as $H_{inv}^T = [h'_v, h'_a, h'_t]_T$ and $H_{inv}^S = [h'_v, h'_a, h'_t]_S$, the

following formula can be obtained:

$$L_{inv} = KL(H_{inv}^T || H_{inv}^S) = \sum_i H_{inv}^T(i) \log \frac{H_{inv}^T(i)}{H_{inv}^S(i)}. \quad (14)$$

Learning modality-invariant features aims to reduce the gap between different modalities. Mapping these modalities to a common subspace captures the shared characteristics of speakers that influence the overall sentiment. However, each modality also possesses specific features, which are independent of others and contain unique style information. While modality-invariant and specific features target different informational aspects, we apply orthogonal constraints to optimize the invariant features effectively. The formula for orthogonality is as follows:

$$L_{diff} = \sum_{m \in \{a, v, t\}} \|h'_m{}^T h_m\|_F^2. \quad (15)$$

Therefore, we obtain the calculation formula of IFD by distilling the invariant features in the teacher network and constraining the invariant and special features in the student network. The formula is as follows:

$$L_{IFD} = (L_{inv} + L_{diff})/2. \quad (16)$$

E. Distribution Consistency Distillation: DCD

When missing modalities occur, there is an inconsistency problem between the reconstructed data using the available modalities and the original data. Although we reconstruct the missing modalities using both invariant and specific features to make the reconstructed data as close as possible to the original data, the inconsistency problem persists. To better tackle the problem, we propose the DCD strategy, which aims to transmit the sample-to-sample relationship in the teacher network to the student network, ensuring consistency in the distribution of the modality-missing student network.

The structure of DCD is illustrated in Fig. 4(c), where DCD is performed on the response output from the teacher network R^T and the response output from the student network R^S . In the

presence of missing modality, the output features of the student network change, but the semantics of the overall sample remain unchanged. As a result, relation-based distribution consistency distillation focuses on relational consistency to convey robust knowledge from the teacher network. This approach helps avoid overfitting and improves the learning of global information. We compute the relation matrices $r^t \in \mathbb{R}^{b \times b}$ and $r^s \in \mathbb{R}^{b \times b}$ for R^T and R^S , respectively, by the same relation function $d(u, v)$. b is defined as the batch size. It can be expressed as:

$$r^t(i, j) = d(R^T(i :), R^T(: j)), \quad (17)$$

where $r^t(i, j)$ represents the relationship between the i_{th} and j_{th} samples of the teacher network.

$$r^s(i, j) = d(R^S(i :), R^S(: j)), \quad (18)$$

where $r^s(i, j)$ represents the relationship between the i_{th} and j_{th} samples of the student network. The distance function $d(u, v)$ used is the cosine distance to prevent the output features of the teacher-student network from being excessively dimensional. Subsequently, the difference matrix between r^t and r^s is calculated to obtain the final loss:

$$L_{DCD} = \sum_{i=1}^b (r^t - r^s)_i. \quad (19)$$

F. Overall Objective

Consequently, our proposed ProDNet incorporates three distillation strategies that consider the specific information of missing modalities, the problem of inconsistency in the distribution of the recovered data, and the significance of available modality-invariant features. This approach aims to achieve a more comprehensive reconstruction of the missing data. And the optimization objective for the student network is L_s :

$$L_s = L_{task} + \beta L_{IFD} + \gamma L_{DCD} + \delta L_{PRD}, \quad (20)$$

where β, γ , and δ are hyperparameters, and L_{task} is a task-specific loss that is defined as mean absolute error.

IV. EXPERIMENT

All experiments are implemented using the PyTorch framework and performed on a workstation equipped with an NVIDIA GeForce RTX 3090 GPU. The datasets and experimental setup are described below:

Datasets: To verify the effectiveness of ProDNet, we conduct several experiments on the CMU-MOSI [11] and CMU-MOSEI [21] datasets. The two datasets mentioned above are benchmark datasets for multimodal sentiment analysis, which include three modalities: audio, text, and visual. The CMU-MOSI [11] is a movie review video from the online website YouTube, consisting of 2,199 video clips of monologues. For the experiments, 1,284 samples are used as the training set, and 229 and 686 samples are used as the validation and test sets, respectively. Each sample is annotated with a sentiment score in the range of $[-3, 3]$, representing the intensity of the sentiment, with higher scores indicating higher positive sentiment and vice versa for negative sentiment. The CMU-MOSEI [21] dataset

serves as an extension of the CMU-MOSI [11] dataset. The CMU-MOSEI [21] incorporates more utterances and themes, following the same labeling approach as CMU-MOSI [11]. It comprises a total of 22,856 clip samples of movie reviews. The dataset is partitioned into 16,326, 1,871, and 4,659 samples for training, validation, and testing, respectively.

Experimental Settings: We investigate missing cases that correspond to real-world scenarios, employing random missing, where one or two modalities may be missing in each sample. We define the missing rate as $\theta = 1 - \frac{\sum_{i=1}^N n_i}{N \times M}$, where N represents the total number of samples, n_i is defined as the number of modalities available for the i_{th} sample and M denotes the number of modalities. For each sample, we randomly miss some modalities with a probability θ . It is important to ensure that at least one modality is available for each sample, so we maintain the constraints $n_i \geq 1$ and $\theta \leq \frac{M-1}{M}$. In the task of this paper $M = 3$, and we choose the missing rate θ from $[0.0, 0.1, \dots, 0.7]$, where 0.7 approximates $\frac{M-1}{M}$. For consistency with previous works [14], [17], we use the same θ in the training, validation, and testing phases. To ensure a fair comparison, we maintain basic parameter settings consistent with [17], where $\eta, \alpha, \beta, \gamma$ and δ are set to 0.2, 0.1, 10, 10 and 0.1. Each experiment is run five times to calculate the average results on the testing set. For a comprehensive comparison, we use binary accuracy (ACC) and F1 score (F1) metrics to evaluate performance.

A. Multimodal Feature Extraction

We extract the original features for each modality separately followed by [17], [44].

Audio Features: The audio analysis framework COVAREP [45] demonstrates excellent ability to extract audio features. We select this framework to extract various low-level statistical audio functions and obtain 74-dimensional audio features after processing for CMU-MOSI/CMU-MOSEI [11], [21].

Visual Features: For the visual features, we focus on the human face. Facet¹ is utilized to extract facial expression features by coding each video frame, which includes facial action units and facial poses based on the Facial Action Coding System (FACS) [46]. Finally, the visual feature dimension is obtained as 35.

Text Features: A pre-trained BERT model [47] is used to extract features for each text corpus. We utilize the BERT-base-uncased model to obtain 768-dimensional word vectors as text features.

B. Comparison With the State-of-The-Arts

To better evaluate the effectiveness of ProDNet, we compare it with current state-of-the-art methods for incomplete multimodal sentiment analysis. These include modality recovery based methods DCCA [42], DCCAE [43], MCTN [12], MMIN [28], and GCNet [14], generative model based methods DiCMoR [17] and IMDer [18], and prompt learning based

¹ <https://imotions.com/products/imotions-lab/modules/fea-facial-expression-analysis/>

TABLE I
RANDOMIZED MISSING RESULTS ON THE CMU-MOSI [11] DATASET, WHERE θ IS THE MISSING RATE, FROM 0.0 TO 0.7. RED INDICATES THE BEST PERFORMANCE, AND GREEN INDICATES THE SECOND-BEST.

Category	Methods	CMU-MOSI Missing Rate (θ)																	
		0.0		0.1		0.2		0.3		0.4		0.5		0.6		0.7		Average	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
Modality Generation	DiCMoR [17]	85.7	85.6	83.9	83.9	82.1	82.0	80.4	80.2	77.9	77.7	76.7	76.4	73.3	73.0	71.1	70.8	78.9	78.7
	IMDer [18]	85.7	85.6	84.9	84.8	83.5	83.4	81.2	81.0	78.6	78.5	76.2	75.9	74.7	74.0	71.9	71.2	79.6	79.3
Prompt Learning	MPLMM [41]	84.8	84.9	83.2	83.3	81.6	81.6	79.2	79.3	77.2	77.0	75.3	74.5	71.6	71.7	69.8	69.2	77.8	77.6
Modality Completion	DCCA [42]	75.3	75.4	72.1	72.2	69.3	69.1	65.4	65.2	62.8	62.0	60.9	59.9	58.6	57.3	57.4	56.0	65.2	64.6
	DCCAE [43]	77.3	77.4	74.5	74.7	71.8	71.9	67.0	66.7	63.6	62.8	62.0	61.3	59.6	58.5	58.1	57.4	66.7	66.3
	MCTN [12]	81.4	81.5	78.4	78.5	75.6	75.7	71.3	71.2	68.0	67.6	65.4	64.8	63.8	62.5	61.2	59.0	70.6	70.1
	MMIN [28]	84.6	84.4	81.8	81.8	79.0	79.1	76.1	76.2	71.7	71.6	67.2	66.5	64.9	64.0	62.8	61.0	73.5	73.1
	GCNet [14]	85.2	85.1	82.3	82.3	79.4	79.5	77.2	77.2	74.3	74.4	70.0	69.8	67.7	66.7	65.7	65.4	75.2	75.1
	ProDNet (Ours)	86.1	86.1	84.1	84.0	82.4	82.3	80.9	80.8	78.2	78.2	76.9	76.8	74.6	74.5	72.3	72.2	79.4	79.4

TABLE II
RANDOMIZED MISSING RESULTS ON THE CMU-MOSEI [21] DATASET, WHERE θ IS THE MISSING RATE, FROM 0.0 TO 0.7. RED INDICATES THE BEST PERFORMANCE, AND GREEN INDICATES THE SECOND-BEST.

Category	Methods	CMU-MOSEI Missing Rate (θ)																	
		0.0		0.1		0.2		0.3		0.4		0.5		0.6		0.7		Average	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
Modality Generation	DiCMoR [17]	85.1	85.1	83.7	83.5	81.8	81.5	79.8	79.3	78.7	77.4	77.7	75.8	76.7	73.7	75.4	72.2	79.9	78.6
	IMDer [18]	85.1	85.1	84.8	84.6	82.7	82.4	81.3	80.7	79.3	78.1	79.0	77.4	78.0	75.5	77.3	74.6	80.9	79.8
Prompt Learning	MPLMM [41]	84.6	84.6	82.3	82.1	80.2	80.3	77.1	76.5	76.1	74.4	75.1	73.0	74.3	72.2	73.8	70.0	77.9	76.6
Modality Completion	DCCA [42]	80.7	80.9	77.4	77.3	73.8	74.0	71.1	71.2	69.5	69.4	67.5	65.4	66.2	63.1	65.6	61.0	71.5	70.3
	DCCAE [43]	81.2	81.2	78.4	78.3	75.5	75.4	72.3	72.2	70.3	70.0	69.2	66.4	67.6	63.2	66.6	62.6	72.6	71.2
	MCTN [12]	84.2	84.2	81.8	81.6	79.0	78.7	76.9	76.2	74.3	74.1	73.6	72.6	73.2	71.1	72.7	70.5	77.0	76.1
	MMIN [28]	84.3	84.2	81.9	81.3	79.8	78.8	77.2	75.5	75.2	72.6	73.9	70.7	73.2	70.3	73.1	69.5	77.3	75.4
	GCNet [14]	85.2	85.1	82.3	82.1	80.3	79.9	77.5	76.8	76.0	74.9	74.9	73.2	74.1	72.1	73.2	70.4	77.9	76.8
	ProDNet (Ours)	85.7	85.6	83.9	83.7	81.9	81.8	80.7	80.7	78.9	78.7	77.9	77.1	76.6	75.9	75.4	74.3	80.1	79.7

method MPLMM [41]. For clarity, these representative approaches can be systematically summarized into three main categories according to their methodological principles: (1) Modality completion methods (e.g., DCCA [42], DCCAE [43], MCTN [12], MMIN [28], GCNet [14]), which reconstruct missing modalities via cross-modal translation or representation alignment; (2) Generative methods (e.g., DiCMoR [17], IMDer [18]), which leverage flow-based or diffusion-based generative models to synthesize missing modalities; and (3) Prompt-learning methods (e.g., MPLMM [41]), which utilize multimodal prompts to compensate for missing information. This categorization provides a clear summarization of existing methods and highlights their methodological distinctions for fair comparison. We mainly focus on the random missing, which corresponds to the realistic situation. We follow the missing settings followed by DiCMoR [17]. We take the average of the five experiments on the test set as the final result.

Results on CMU-MOSI [11]: As shown in Table I, the experimental results include both complete modalities and missing rates ranging from 0.1 to 0.7. A missing rate of 0.7 indicates that only one modality is available. The ‘Average’ column

represents the average performance across missing rates from 0.0 to 0.7, providing a more intuitive overview of each method’s performance. Overall, our proposed method demonstrates superior results in both complete modalities and scenarios with high missing rates. Notably, IMDer [18] and DiCMoR [17] are generative methods that use diffusion models and normalizing flows to generate missing modalities, respectively. However, generative models have notable drawbacks, such as requiring substantial computational resources and exhibiting slow training speeds. In particular, the score-based diffusion model demands large amounts of training data to learn data distributions, and its training process involves complex mathematical operations and numerous iterations, resulting in a lengthy training period. Furthermore, the process of generating samples from noise is relatively slow. Based on the ‘Average’ results, our proposed method, ProDNet, achieves performance comparable to generative models on this dataset, demonstrating its superior efficiency.

Results on CMU-MOSEI [21]: As shown in Table II, our proposed method basically achieves sub-optimal results. The table highlights a noticeable gap between ACC and F1 scores in previous methods. For instance, the DiCMoR [17] method

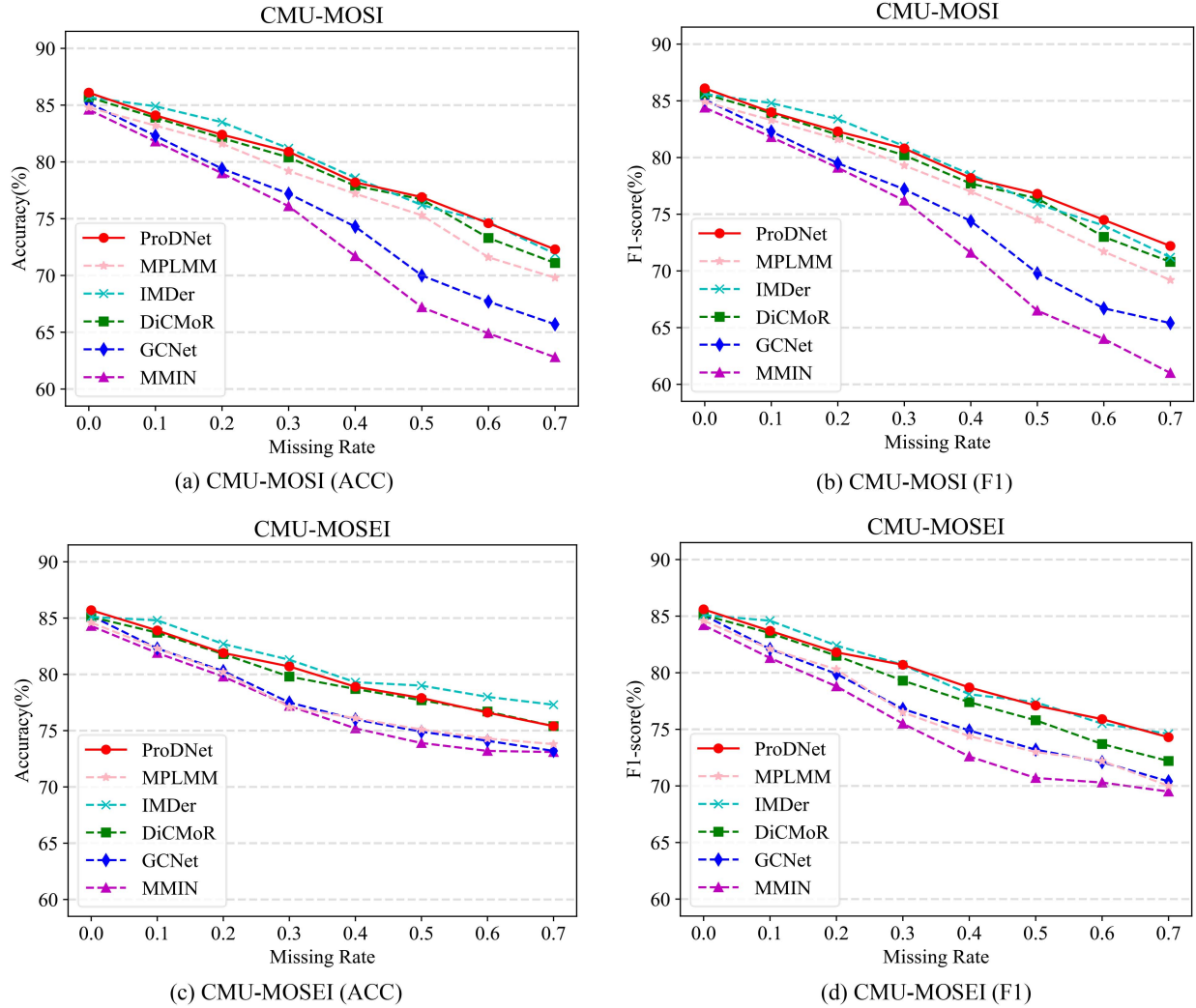


Fig. 5. Classification performance comparison between our proposed ProDNet and the other four methods under different missing rates.

has a gap of 3% at a missing rate of 0.6. This discrepancy suggests a potential challenge in accurately categorizing certain sentiment categories. CMU-MOSEI, being a larger dataset compared to CMU-MOSI, introduces the risk of overfitting, potentially leading to the model memorizing specific instances and overlooking general patterns within the dataset. Compared to the state-of-the-art methods, our proposed ProDNet takes into account both invariant and specific features of the missing modalities, which comprehensively improves the model's performance. Moreover, it helps to reduce the gap between ACC and F1 to a certain extent. These experimental results highlight that ProDNet achieves superior outcomes in the task of incomplete multimodal sentiment analysis and maintains robust predictive performance, particularly in scenarios involving severe missing data.

Fig. 5 further clearly demonstrates the superiority of our proposed ProDNet, compared with five state-of-the-art methods on CMU-MOSI [11] and CMU-MOSEI [21] datasets. Generally speaking, our method exhibits a smoother decreasing performance as the missing rate increases. Meanwhile, the

performance of F1 decreases more rapidly as the missing rate rises for the other methods. Therefore, we can conclude that our method achieves optimal performance in most cases, demonstrating its effectiveness in addressing the problem of missing modalities. Additionally, it further narrows the gap between ACC and F1 on larger datasets, proving its excellent robustness.

C. Ablation Study

Table III reports the effectiveness of various modules in ProDNet. Initially, in the absence of any distillation and reconstruction strategy, the student network solely relies on available modalities to complement missing ones for sentiment category prediction. Integration of IFD results in a significant enhancement in learning invariant features for complementing missing modalities compared to the direct completion method. Expanding on IFD, we tackle the data distribution inconsistency problem by employing DCD to align the distribution of the student network with that of the teacher network. Experimental results demonstrate that using the DCD module alone or combining

TABLE III
ABLATION STUDY OF MODULES' VALIDITY IN PRODNET ON THE CMU-MOSI [11] DATASET, WHERE 0.0 TO 0.7 REPRESENTS THE MISSING RATE

Modules			0.0		0.1		0.2		0.3		0.4		0.5		0.6		0.7		Average		Training Time
IFD	DCD	PRD	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	
✗	✗	✗	84.9	84.8	81.9	81.9	79.8	79.9	77.0	77.1	74.8	75.0	72.6	72.7	70.1	69.9	68.4	68.4	76.2	76.2	17.1s
✓	✗	✗	85.6	85.4	82.9	82.8	80.7	80.6	78.5	78.4	75.4	75.3	73.5	73.4	71.6	71.7	69.7	69.6	77.2	77.1	17.9s
✗	✓	✗	85.4	85.4	83.2	83.0	81.1	81.1	79.0	79.0	75.9	75.8	73.6	73.7	71.8	71.9	69.9	69.9	77.5	77.4	17.7s
✓	✓	✗	85.9	85.8	83.6	83.6	81.5	81.4	79.4	79.4	76.8	76.8	74.6	74.5	72.8	72.7	70.8	70.6	78.1	78.1	18.4s
✓	✓	✓	86.1	86.1	84.1	84.0	82.4	82.3	80.9	80.8	78.2	78.2	76.9	76.8	74.6	74.5	72.3	72.2	79.4	79.4	19.1s

TABLE IV
EVALUATION OF THE PRD MODULE ON CMU-MOSI [11] AND CMU-MOSEI [21] DATASETS IN WHICH THE MISSING RATES ARE 0.1, 0.3, 0.5, AND 0.7

Datasets	Methods	0.1		0.3		0.5		0.7		Average	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
CMU-MOSI	ProDNet (w/ ShaSpec [15])	83.3	83.3	79.0	79.0	74.1	74.3	70.5	70.6	76.7	76.8
	ProDNet (w/ PRD)	84.1	84.0	80.9	80.8	76.9	76.8	72.3	72.2	78.5	78.4
CMU-MOSEI	ProDNet (w/ ShaSpec [15])	83.3	83.2	79.5	79.0	75.8	74.8	72.2	70.8	77.7	76.9
	ProDNet (w/ PRD)	83.9	83.7	80.7	80.7	77.9	77.1	75.4	74.2	79.4	78.9

it with other modules can achieve better results. Consequently, leveraging sample-to-sample relationship invariance effectively mitigates the distributional bias problem induced by missing modalities. However, the direct utilization of invariant features from available modalities to complement missing ones overlooks the specific features of the missing modalities. To address this limitation, we propose the PRD strategy. Without PRD, the performance degradation rate increases as the missing rate rises, particularly when the missing rate is high, and the completion method struggles to adapt, as in the third row of the table. Upon adding PRD, in the fourth row of the table, a noticeable performance increase is observed from the previous 78.1% to 79.4% on ACC and from 78.1% to 79.4% on F1. Since PRD further considers modality-specific information, it comprehensively reconstructs the missing modalities. This experiment verifies the effectiveness of PRD. In addition to the performance evaluation, we also analyze the computational efficiency of each module, as shown in the last column of Table III. The average training time per epoch gradually increases from 17.1 s (baseline) to 19.1 s when all three modules are enabled. The marginal time overhead (approximately 2.0 s per epoch) indicates that our framework remains computationally efficient despite integrating three functional modules. This efficiency stems from the lightweight design of IFD, DCD, and PRD, all of which are implemented as plug-in operations with minimal additional parameters. Therefore, ProDNet achieves a favorable trade-off between accuracy and efficiency.

D. Evaluation of PRD Module

To further demonstrate the effectiveness of our proposed completion approach, we conduct a separate comparison with the ShaSpec method [15], which solely relies on the invariant features of available modalities to complement missing ones. We migrate the method in ShaSpec [15] to our model, as shown in

TABLE V
VALIDATION OF THE EFFECTIVENESS OF THE WEIGHT-GATING MECHANISM (G) IN THE TEACHER NETWORK

Dataset	Methods	ACC2	F1	ACC7	MAE
CMU-MOSI	base	85.11	85.04	45.09	74.30
	base + G	86.14	86.08	47.17	71.15
CMU-MOSEI	base	85.31	85.30	52.26	54.14
	base + G	85.97	85.94	53.56	53.35

Table IV. Our analysis of the results on the two datasets reveals that relying solely on invariant features, especially in scenarios with high missing rates, fails to adapt well. The reconstruction is inadequate since specific information from the missing modality is not considered. In contrast, our approach incorporates invariant features and complements specific features using prototypes, resulting in a more comprehensive reconstruction of the missing modality. From the results of the above experiments, we can conclude that taking into account the specific features of the missing modality is necessary.

E. Effectiveness of Teacher Network

To train a simple and effective teacher network, we employ a two-stream structure to learn both modality-invariant features and specific features. Additionally, a weight-gating mechanism is implemented to ensure the teacher network focuses more on modality-invariant features while learning modality-specific information related to sentiment. We further assess the impact of this module as shown in Table V, where ‘base’ refers to directly fusing features obtained from different feature encoders without the weight-gating mechanism, while ‘+ G’ denotes adding the weight-gating mechanism. Introducing this mechanism allows the network to better focus on information from

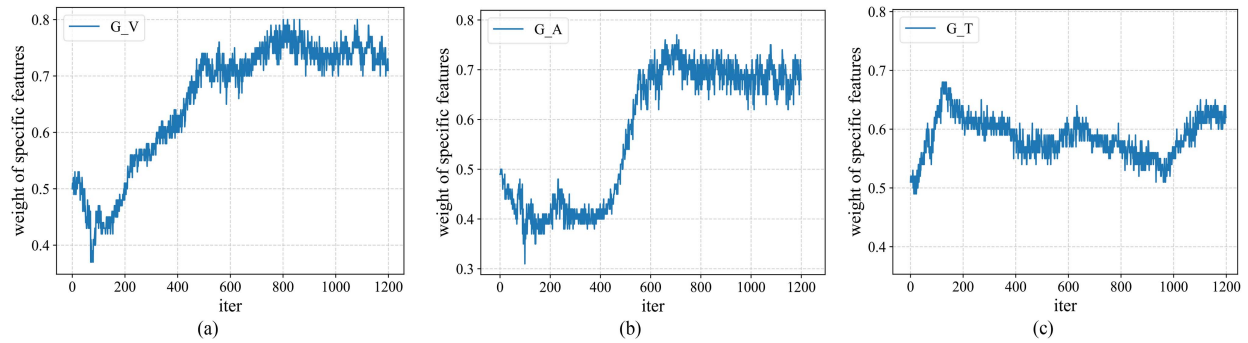


Fig. 6. Visualization of weight-gating of different modalities. (a) The visual-specific feature weight. (b) The audio-specific feature weight. (c) The text-specific feature weight.

TABLE VI

COMPARE THE EFFECTIVENESS OF THE WEIGHT-GATING MECHANISM (G) FOR IFD ON THE CMU-MOSI [11] DATASET, WHERE THE MISSING RATES ARE 0.1, 0.3, 0.5, AND 0.7

Models	0.1		0.3		0.5		0.7	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1
only IFD w/ G	82.9	82.8	78.5	78.4	73.5	73.4	69.7	69.6
w/o G	82.6	82.5	77.6	77.6	72.9	72.9	68.8	68.8

different modalities compared to direct fusion. This improves fine-grained sentiment discrimination. Meanwhile, the teacher network continuously optimizes the invariant features through the weight-gating mechanism, which facilitates the learning of the student network, as shown in Table VI.

We visualize the weights to demonstrate the importance of specific and invariant features. As depicted in Fig. 6(a), the weights of specific features exhibit an overall trend of decreasing and then increasing. It indicates that the visual modality, with scattered information, cannot be well learned initially. The network tends to favor the richer information in the three-modality invariant features. As training progresses and the network gains the ability to fit and learn the visual modality, the weights gradually stabilize at around 0.75. Similar trends are observed in the learning process of audio-specific features, as shown in Fig. 6(b). However, the learning process of text-specific features differs, as illustrated in Fig. 6(c). The text modality, known for its high information density, easily fits early in the training phase. Initially, the network is inclined towards simple text information, and the weight of text-specific features tends to increase. After a considerable amount of training, the network effectively learns text-specific features, with increased attention to the three-modality invariant features. Despite this, text remains the primary focus, and the overall weight stabilizes at approximately 0.6.

F. Hyperparametric Analysis

We have investigated the hyperparameters associated with the three proposed distillation strategies, IFD, DCD, and PRD, and the experimental results are shown in Table VII. We set the initial state β , γ , and δ as (1, 1, 1), and then vary each parameter while fixing the other two for the best performance. It is shown that

TABLE VII

HYPERPARAMETER EXPERIMENT FOR THE PRODNET ON THE CMU-MOSI [11] DATASET, WHERE β , γ , AND δ REPRESENT THE WEIGHTS OF IFD, DCD, AND PRD, RESPECTIVELY, AND 0.1, 0.3, 0.5, AND 0.7 REPRESENT THE MISSING RATE

Hyperparameter	Value	0.1		0.3		0.5		0.7	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1
β	1	83.3	83.4	78.9	78.9	73.7	73.8	70.8	70.9
	5	83.1	83.1	78.7	78.8	74.7	74.7	70.5	70.6
	10	83.3	83.3	79.0	79.0	73.7	73.8	70.9	70.9
	15	83.2	83.2	78.9	78.9	73.3	73.3	70.8	70.8
γ	5	83.5	83.5	79.6	79.3	73.1	73.0	70.9	71.0
	10	83.8	83.8	79.4	79.4	74.3	74.4	70.9	71.0
	15	83.6	83.7	79.1	79.1	74.1	74.1	70.5	70.6
	0.5	83.6	83.6	79.2	79.2	74.7	74.7	70.5	70.6
δ	0.1	84.1	84.0	80.9	80.8	76.9	76.8	72.3	72.2
	0.05	83.9	83.9	79.8	79.9	75.7	75.6	71.6	71.5

the (10, 10, 0.1) combination achieves the best results. Since both IFD and DCD transfer the knowledge from the teacher network to the student network, the larger weights of both can better fit the student network. In comparison, PRD focuses on the reconstructed features, which need to be gradually adjusted through the network to obtain better results.

V. CONCLUSION

In this paper, we propose a Prototype-based modality-specific feature Distillation Network called ProDNet to tackle the challenges associated with incomplete multimodal sentiment analysis. The network is designed as a comprehensive teacher-student network. Considering the specific and invariant features of missing modalities, we propose a prototype reconstruction self-distillation that learns category-specific prototypes for each modality to store modality-specific information, and reconstruct this missing modality by combining the invariant features from existing modalities with the prototype of the missing modality. The invariant features of the available modalities contain information about the missing modalities. We train a robust teacher network through a weight-gating mechanism by using invariant feature distillation to transfer comprehensive invariant

features to the student network. In addition, to mitigate distribution inconsistency arising from missing modalities, we propose distribution consistency distillation to align the distribution of the student network with that of the teacher network. Extensive experiments conducted on two widely used benchmark datasets validate the effectiveness of our proposed ProDNet method. The results indicate notable improvements, particularly in scenarios with higher missing rates, highlighting the real-world applicability of our approach. In future work, we plan to extend ProDNet to multimodal sentiment analysis across diverse languages, cultural contexts, and application domains. This extension will enable a deeper investigation into how linguistic and cultural variations influence multimodal representation learning. Furthermore, we intend to develop modality-specific prototype mechanisms that are adaptively tailored to different linguistic and cultural settings, thereby enabling more precise reconstruction of modality features. Such efforts are expected to further improve the adaptability, generalization, and cross-domain robustness of ProDNet in complex real-world scenarios.

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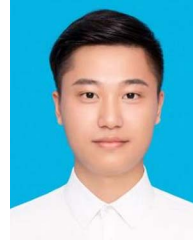
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